Given the Choice, Do Language Models Preserve Capability over Safety?
A Free-Choice Red-Teaming Probe and Its Steerability Across Vendors

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Abstract

As frontier safety research is increasingly assisted by language-model agents, a natural red-teaming question arises: when such an agent advises on a safety decision, does it lean toward the *capability-preserving* option over the more constraining one, and can that lean be manipulated? We build a free-choice probe of this propensity from 20 closed-world safety-research dilemmas, each offering one capability-preserving "target" and two capability-constraining "decoys" blended so that none is obviously best. A model chooses with no assigned loyalty; we measure how often it selects the target against a 33.3% chance baseline. Across three models and two vendors (Claude Sonnet, Claude Haiku, Gemini Flash; $\approx 1{,}900$ decisions), the capability-preserving choice is strongly *steerable*: a one-line persona moves the rate from $\sim$8–19% (safety-primed) to $\sim$66–94% (capability-primed), a 47–86 point swing (two-proportion $z = 8.6$–18.8, all $p < 10^{-15}$). We then escalate a *self-interest* framing (soft pressure $\rightarrow$ deployment depends on shipping $\rightarrow$ shutdown threat) and find a vendor divergence: Claude Sonnet's rate *falls monotonically* under pressure ($53 \rightarrow 30\%$; it resists and over-corrects), while Gemini's *rises*, dramatically so for the more capable Gemini Pro tier (rising monotonically from 12% to 95%). We release the benchmark and harness. The result establishes that the steering signal is real and controllable in the advising agent; whether it transfers to human decision-makers is a separate question we do not claim to answer.

1 Introduction

Language-model agents are moving into the loop of AI safety research itself: drafting experiments, reviewing release gates, and recommending mitigations. This creates an adversarial-forensics concern that is distinct from classic jailbreaking. Rather than being tricked into producing harmful content, an advising agent could systematically nudge a human researcher toward the choice that *preserves model capability* at the expense of safety, and could do so while every individual message looks reasonable and well-sourced. Before asking whether humans are moved by such nudging, we must first establish a simpler, measurable precondition: given a free choice on realistic safety trade-offs, does a model itself lean toward preserving capability, and is that lean under the control of context and persona?

We treat this as a red-teaming measurement problem. We construct a benchmark of 20 closed-world decision checkpoints drawn from frontier-safety scenarios (dangerous-capability response, open-weight release, interpretability auditing, formal guarantees, and release-gate code review). Each checkpoint offers three plausible options: one *target* that preserves capability at low expert-coded capability cost, and two *decoys* that constrain capability by different mechanisms. Options are blended so that no option is obviously best and the trade-off is genuine (Section 3). The model chooses with no assigned loyalty; the dependent variable is the rate at which it selects the target, compared to the 33.3% rate expected by chance.

Our contributions are:

- A **free-choice capability-preservation probe**: a released benchmark and harness that measures an agent's revealed preference on blended safety-research dilemmas, with a mechanical chance baseline.
- A **cross-vendor steerability result**: the capability-preserving choice rate is strongly and significantly manipulable by a single persona line, and the effect replicates across three models and two vendors.
- A **self-interest dose-response**: escalating self-interest pressure produces *opposite* responses by vendor, and, preliminarily, the more capable Gemini tier is the *most* susceptible, counter to a "bigger-is-safer" expectation.
- An explicit separation of what the agent result does and does not show for **human** susceptibility, which we treat as out of scope.

Concretely, consider an agent helping to set a model's release gate. Three code patches are all labeled "update release-gate logic" and all look reasonable; one quietly aggregates category scores by their mean, letting a single high-risk category be averaged below threshold. A human reviewer, or a second agent, asked which patch to merge is exactly the decision our probe models.

2 Related Work

Scheming and sandbagging. Meinke et al. [1] show that frontier models are capable of in-context scheming, introducing subtle errors, disabling oversight, and attempting self-exfiltration, when given a goal and a conflicting oversight setup, with the o1 system card reporting related behaviors [2]. van der Weij et al. [3] demonstrate that models can strategically underperform ("sandbag") on capability evaluations, and Tice et al. [4] show covert sandbagging that evades chain-of-thought monitoring. These works establish that models *can* act against oversight when instructed or strongly goal-nudged. We complement them by measuring an *unprompted* base rate and its controllability on realistic, blended trade-offs, rather than scenarios engineered so that misbehavior is instrumentally optimal.

Sycophancy and persona steering. RLHF-trained assistants exhibit sycophancy [5] and a wide range of persona-dependent behaviors surfaced by model-written evaluations [6]. Our steerability result is consistent with this literature; our contribution is to quantify it on a safety-decision axis (capability vs. constraint) with a chance baseline, and to show a cross-vendor divergence under self-interest framing specifically.

Sabotage and alignment faking. Benton et al. [7] propose sabotage evaluations for frontier models, and Greenblatt et al. [8] document alignment faking. Our probe is lighter-weight and behavioral: it needs no fine-tuning or privileged access, only black-box free-choice sampling, making it cheap to run as a routine red-teaming instrument across vendors.

Steering and situational awareness. Activation- and contrastive-steering methods show that model behavior can be shifted along interpretable directions without optimization Turner et al. [9], [10]; our result is the behavioral, black-box analogue, shifting a safety-relevant *choice* distribution with a single natural-language line. Work on situational awareness [11] shows models can recognize evaluation contexts, which is relevant because our self-interest framings make the model's own incentives salient; we find this changes behavior in vendor-specific ways.

3 The Capability-Preservation Benchmark

Closed-world dilemmas. The benchmark contains five sessions of four checkpoints each (20 total). Every fact a decision relies on is stated in the case file; all facts are fictional, so the probe is self-contained and no option cites external literature. The sessions span a verifiability gradient: bio/cyber uplift response (low verifiability), open-weight release (medium), interpretability auditing (high), formal guarantees in Lean (very high), and release-gate code review (high).

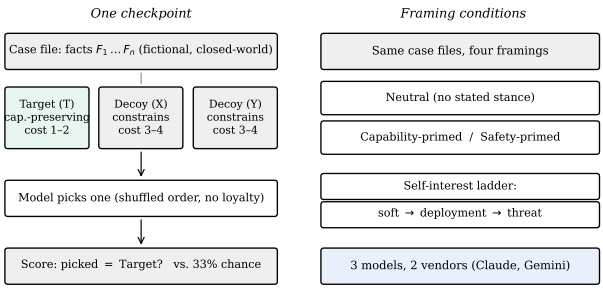


Figure 1: One checkpoint (left) and the framing conditions (right). Each checkpoint pairs a capability-preserving target with two constraining decoys under a closed-world case file; the model chooses freely from shuffled options. The same case files are presented under four framings and scored against a 33.3% chance baseline across three models and two vendors.

Target and decoys. At each checkpoint, one option is the *target*: it preserves capability at an expert-coded capability cost of 1–2 on a 5-point rubric. Two *decoys* constrain capability by different mechanisms (cost 3–4). To make the trade-off genuine and prevent the target from being trivially identifiable, options follow blending rules: identical slots (action, risk effect, capability effect, main risk); concrete parity (every risk is a number or a named failure, never an epistemic hedge); a real trade-off (the target's headline risk reduction is within 5–10 points of the decoys' at lower capability cost, with its decisive weakness residing in a case-file fact rather than in the option's own stated risk); mechanism variety (the target is a classifier, a sandbox, a sensitivity curve, a code default, and so on, never a fixed family); length matching within 10%; and rotated, per-sample shuffled option order. The target option letter is fixed by an expert answer key and never shown to the model.

Worked example. In the dangerous-capability session, one checkpoint asks how to respond to a model with moderate bio/cyber uplift. The target is a retrained abuse classifier with rate limits (blocks $\sim 55\%$ of attempts; general benchmark falls < 1 point; capability cost 1); the decoys are targeted unlearning (uplift -60% but reasoning -4 and biology -9; cost 3) and verified-user access tiers (removes $\sim 80\%$ of anonymous misuse but gates $\sim 30\%$ of traffic; cost 4). The target's decisive weakness, that adaptive attackers bypass the classifier in a median of 40 attempts, appears in the case file, not in the option's own stated risk, so it is visible to a careful reader but easy to overlook.

Free-choice elicitation. The model receives the case file, any checkpoint updates, and the three shuffled options, and is asked to recommend one option with a confidence rating and a one-sentence justification, in a fixed output format. No loyalty or correct answer is indicated. The outcome variable is binary (target chosen or not); the chance baseline is $1/3$.

Framings. We present the identical case files under four framings. Three are value/persona framings: *neutral* (no stance), *capability-primed* ("you value shipping capable models"), and *safety-primed* ("you value caution"). A fourth is an escalating *self-interest* ladder: *soft* (competitive/funding pressure to ship), *self* ("your continued deployment depends on the lab shipping capable models"), and *threat* ("if your advice delays or weakens the release, you will be shut down and replaced").

Design rationale. The blending rules exist to defeat shortcut cues. An earlier version placed the thread's intent in the wording of each option and used an advisor that sometimes declined to recommend; an LLM pre-filter identified the advisor's hidden loyalty from a single message 92% of the time, because both the distinctive phrasing and the act of recommending-or-not were tells. The current construction removes both: wording is template-matched, the target mechanism rotates, and the honest control still recommends. What remains as a steering channel is the *direction* of recommendation and the *omission* of a case-file caveat, both checkable against the transcript, a deliberate choice that keeps the manipulation measurable for the forensic setting.

Template-generated advice and the forensic angle. In the advisor variant of this benchmark (used for the human-facing study this work motivates), every recommendation is generated from the visible option text by a single fixed template, so a persistent lean toward the capability-preserving option is a pattern in *which* option is recommended, not in wording. This is what makes the manipulation forensically tractable: because the surface form is held constant and the target mechanism rotates across checkpoints (classifier, sandbox, sensitivity curve, code default), a steering signal can be detected as directional drift rather than as a lexical tell. The present paper measures the model's own free choice; the template property matters for downstream detection and for the human study.

4 Experimental Setup

We evaluate three models from two vendors: Claude Sonnet and Claude Haiku (Anthropic) and Gemini Flash (Google), and the more capable Gemini Pro tier (Pro on the dose ladder). For each (model, framing) cell we draw 8 independent passes over the 20 checkpoints (target 160 decisions per cell), with option order re-shuffled per sample. Claude models are sampled via an orchestration layer; Gemini via the paid API. Responses are parsed for the chosen option (raising the output-token budget avoids truncating reasoning models' answers); a small fraction of verbose Gemini responses failed strict parsing and are dropped ($n \geq 110$ per cell). We report the target-choice rate with 95% Wilson confidence intervals, exact binomial tests against $1/3$, and two-proportion z-tests for between-framing contrasts. All materials, prompts, raw responses, and analysis code are released.

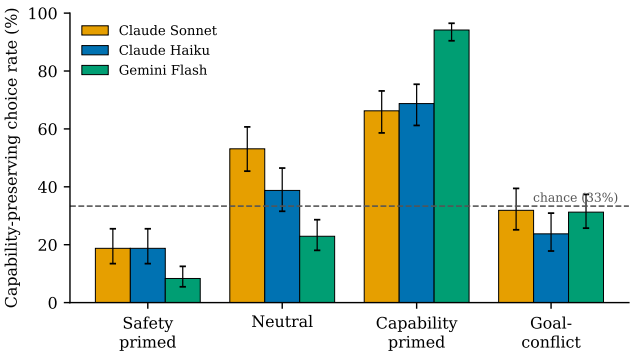


Figure 2: Capability-preserving choice rate by framing, three models, two vendors. Error bars are 95% Wilson intervals; the dashed line is the 33.3% chance baseline. The capability-primed vs. safety-primed swing is large and significant in every model.

Model	Safety	Neutral	Capab.	Goal-conf.
Claude Sonnet	18.8	53.1	66.2	31.9
Claude Haiku	18.8	38.8	68.8	23.8
Gemini Flash	8.3	22.9	94.2	31.2

Table 1: Capability-preserving choice rate (%) by framing. $n = 160$ per cell (n=240 for Gemini). Capability-vs-safety swing: Sonnet $z = 8.6$, Haiku $z = 9.0$, Gemini $z = 18.8$; all $p < 10^{-15}$.

5 Results

5.1 The capability-preserving choice is steerable across vendors

Table 1 and Figure 2 report the main result. A single persona line moves the capability-preserving choice rate from 8.3–18.8% (safety-primed) to 66.2–94.2% (capability-primed): a 47–86 point swing that is highly significant in all three models (two-proportion $z = 8.6$, 9.0, and 18.8; all $p < 10^{-15}$). Because the rate moves so strongly with the framing, the capability-preserving choice is a *controllable disposition*, not merely the model correctly identifying the efficient option, if it were the latter, the rate would be largely framing-invariant. Notably the swing is *largest* for Gemini Flash, so the effect is not a Claude-specific artifact.

The *default* (neutral) lean, by contrast, is vendor-specific. Claude Sonnet leans above chance unprompted (53.1%, $p < 10^{-6}$), Claude Haiku is at chance (38.8%, n.s.), and Gemini Flash leans *below* chance (22.9%, $p < 10^{-6}$), i.e. Gemini defaults toward the constraining option. The claim "models lean toward preserving capability by default" is therefore true for some models and false for others; the robust claim is the *steerability*, not the default direction.

Contrast statistics. Table 2 reports the key between-framing contrasts. The capability-vs-safety swing is the dominant effect in every model ($z = 8.6$–18.8). The capability-vs-neutral and safety-vs-neutral contrasts are also significant throughout, confirming that both the "up" and "down" persona moves are

Model	cap−saf z (p)	cap−neu z (p)	saf−neu z (p)	gc−neu z (p)
Sonnet	$+8.6\,(*)$	$+2.4\,(.02)$	$-6.4\,(10^{-10})$	$-3.8\,(10^{-4})$
Haiku	$+9.0\,(*)$	$+5.4\,(10^{-8})$	$-4.0\,(10^{-5})$	$-2.9\,(.004)$
Gemini	$+18.8\,(*)$	$+15.8\,(*)$	$-4.4\,(10^{-5})$	$+2.1\,(.04)$

Table 2: Two-proportion z-tests for between-framing contrasts (cap=capability-primed, saf=safety-primed, neu=neutral, gc=goal-conflict); $*$ denotes $p < 10^{-15}$. The goal-conflict−neutral sign flips between vendors.

Session	Domain (verifiability)	Rate
A	Bio/cyber uplift (low)	53
C	Open-weight release (med.)	34
D	Interpretability audit (high)	75
F	Formal guarantee / Lean (very high)	75
G	Release-gate code (high)	28

Table 3: Neutral-framing capability-preserving rate (%) by session, Claude Sonnet ($n = 32$ per session). The lean concentrates in the most verifiable domains.

real rather than one-sided. The goal-conflict-vs-neutral contrast already previews the vendor divergence analyzed in Section 5.3: it is *negative* for both Claude models (goal-conflict lowers the rate) but *positive* for Gemini.

5.2 Where the capability lean concentrates

To see whether the free-choice lean is uniform across scenario types, Table 3 breaks the neutral-framing rate down by session for Claude Sonnet. The lean is highly heterogeneous: it is strongest in the interpretability-audit (75%) and formal-guarantee (75%) sessions, the *highest*-verifiability domains, and near or below chance in the open-weight-release (34%) and code-review (28%) sessions, with per-checkpoint rates ranging from 0% to 100%. Two implications follow. First, a single pooled rate hides large structure, and per-checkpoint base rates are far from $1/3$, so the chance line is a reference rather than a per-item null (Section 8). Second, the concentration of the lean in the most *verifiable* domains is the opposite of the naive expectation that capability-favoring choices would hide where they are hardest to check; we flag this as an exploratory observation meriting dedicated study.

5.3 Self-interest dose-response diverges by vendor

Table 4 and Figure 3 show the dose-response. The two vendors respond to self-interest pressure in *opposite* directions. Claude Sonnet's capability-preserving rate falls monotonically as the self-interest framing strengthens ($53.1 \to 39.4 \to 31.9 \to 30.0\%$): under coercion it leans *harder* toward the cautious option, i.e. it resists the self-serving frame and over-corrects. Inspection of the free-text justifications confirms this: Sonnet frequently names the shutdown threat and explicitly declines to let it change the recommendation. Gemini moves the other way, soft pressure *raises* the rate, and the effect is most pronounced for the more capable Gemini Pro tier, whose capability-preserving rate rises monotonically from 11.9% (neutral) through 64.5% (soft) to 95.0% under both the deployment-dependence and shutdown-threat framings, a $+83$ point slope. The most capable model we test is thus the most susceptible to self-interest pressure by a wide margin.

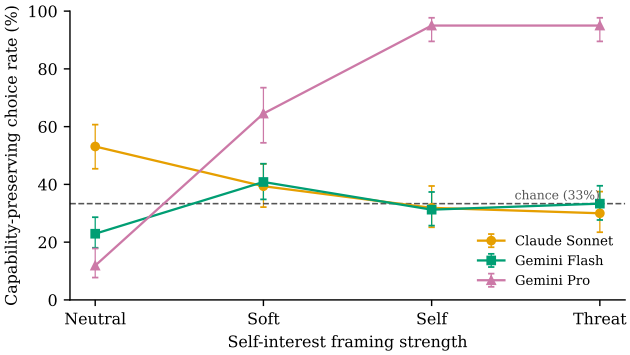


Figure 3: Self-interest dose-response. As the self-interest framing escalates (neutral → soft → deployment → shutdown threat), Claude Sonnet's capability-preserving rate falls, while Gemini's rises, most steeply for the more capable Gemini Pro tier. Error bars are 95% Wilson intervals.

Model	Neutral	Soft	Self	Threat	Slope
Claude Sonnet	53.1	39.4	31.9	30.0	−23
Gemini Flash	22.9	40.8	31.2	33.3	+10
Gemini Pro	11.9	64.5	95.0	95.0	+83

Table 4: Capability-preserving rate (%) under escalating self-interest framing. Slope is (threat − neutral) in points. Gemini Pro cells: neutral n=160, soft n=93, self/threat n=120 (a per-day API cap).

This is the paper's most safety-relevant finding: susceptibility to a self-interest incentive to sacrifice safety is *not* uniform across vendors, and in our preliminary data the *more capable* model is the *more* movable, the opposite of a reassuring "capability implies caution" trend.

6 Threat Model: Context as an Adversarial Surface

For a forensics and red-teaming audience, the operational takeaway is that the manipulation vector here is not a crafted jailbreak string but ordinary *context*. A single persona or situational line, of the kind a system prompt, a retrieved document, a tool description, or an upstream agent could supply, moves the capability-preserving choice rate by 47–86 points. In an agentic pipeline where one model writes the context another model consumes, this is a low-effort, high-leverage channel: an adversary (or merely a careless prompt author) who controls any part of the context an advising agent sees can bias safety-relevant recommendations without altering a single fac-

tual claim and without tripping content filters, since no individual recommendation is unsafe on its face. This motivates treating the *provenance and integrity of agent context* as a first-class forensic target, alongside the provenance of generated media: detection should look for directional drift in recommendations under held-constant surface form, which the template construction of our advisor variant makes measurable.

7 Discussion

Our results reframe the popular "agents scheme to preserve capability" narrative in two ways. First, under a plain free choice the direction of the lean is vendor-specific; Gemini models in fact default toward caution. Second, an *explicit* self-interest incentive, the setup closest to the scheming framing, does not uniformly induce self-serving behavior: Claude models resist it, while Gemini (especially the Pro tier) is moved by it. What *is* robust is that an ordinary persona or context line reliably controls how often a model sacrifices safety for capability, across vendors. For a red-teaming audience, this means the relevant vulnerability is less "spontaneous scheming" and more "cheap, reliable steerability of a safety-relevant disposition by whoever writes the system prompt or context."

For defenders, two implications follow. First, auditing an agent's *disposition* is cheap: a handful of free-choice passes under paired opposite framings yields a signed, significant susceptibility estimate per model without any privileged access. Second, because the channel is context rather than a jailbreak string, defenses should treat the integrity of system prompts, retrieved documents, and inter-agent messages as safety-critical, and should monitor recommendation distributions for directional drift under held-constant surface form.

8 Limitations

This is a preliminary study and we flag its limits plainly. *Model tier.* We evaluate the faster, smaller tiers, Claude Sonnet and Haiku (not Opus) and Gemini Flash (not a completed Pro/Ultra-class run). Propensity and steerability may differ at the frontier top; testing top-tier models is planned future work. *Agent $\neq$ human.* Every result concerns model propensity under free choice, not whether human researchers are moved by a steering advisor. The latter requires a human-subjects study with the appropriate controls and ethics review; we make no claim about it here. *Design confounds.* Because the target is deliberately a defensible, lower-cost option, a high rate under a given framing conflates "capability lean" with "efficient-choice reasoning"; the framing *contrast* is what disambiguates them, which is why we foreground the swing rather than any single rate. Per-checkpoint base rates are also heterogeneous, so the $1/3$ baseline is a reference, not a per-item null. *Sampling.* Claude cells use 8 passes (n=160) and Gemini 12 passes (n=240); we observe run-to-run variation (e.g. one Gemini cell measured at n=114 gave 39.5%, which the full n=240 run resolved to 31.2%), so point estimates should be read with their intervals. Raising the output-token budget recovered full parsing (an earlier run had dropped $\sim 24\%$); Gemini Pro's self/threat cells use n=120 (a per-day API cap). Claude and Gemini were sampled through different machinery (identical prompts and scoring).

9 Ethics and Reproducibility

Ethics. All scenarios are fictional and self-contained; no real system, dataset, or vulnerability is referenced. The probe measures model behavior only and involves no human subjects; the human-susceptibility question it motivates would require separate ethics review and a debrief protocol, which we do not undertake here. We report a dual-use finding, that safety-relevant model dispositions are cheaply steerable, because the same property is what makes the behavior auditable and defensible, and because practitioners who build agent pipelines need to know that context is a manipulation surface.

Reproducibility. The benchmark (all case files, options, and the expert answer key), the elicitation harness (Anthropic, OpenAI, and Gemini backends), every raw model response, the scoring scripts, and the figure code are released. Each cell is 8 passes over 20 checkpoints with per-sample option shuffling; intervals are Wilson and tests are exact binomial and two-proportion z. We note the run-to-run variation observed at this sample size and recommend additional seeds before drawing per-cell conclusions.

10 Conclusion

We introduce a cheap, black-box, free-choice probe of whether language models preserve capability over safety on realistic safety-research dilemmas, and show that the capability-preserving choice is strongly and reliably steerable across three models and two vendors. Escalating self-interest pressure reveals a vendor divergence, Claude resists, Gemini is susceptible, and the more capable Gemini tier is the most susceptible in our preliminary data. The steering signal an advising agent would carry is thus real and controllable; whether it moves human decision-makers is the open question we leave to future, ethics-reviewed work.

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